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Further study needed of the impact of early-life surgery with associated general anesthesia on the risk of attention deficit hyperactivity disorder (ADHD)

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Ing *et al* should be commended for their work in two recent studies published in the *Journal*, in which they made methodologically sound attempts to address the potential impact of surgery with associated general anesthesia on the risk of subsequent diagnoses of, or medication use for, attention deficit hyperactivity disorder (ADHD) and other mental illnesses in 0-4 year-old children from Texas and New York.[1,2]

Together, the two studies strongly suggest that common surgical procedures, which are usually carried out under general anesthesia during infancy or early childhood, such as pyloromyotomy, inguinal hernia repair, postneonatal circumcision and tonsillectomy and/or adenoidectomy, are associated with an increased risk of ADHD. As the authors rightly acknowledge, it is currently not possible from available data to determine whether the association is because of harmful effects of surgery *per se* or/and associated traumatic experiences in the perioperative period or because of the general anesthesia that accompanies most, but not all, of these operations. Further study is needed to clarify this.

Toward this end, we would like to bring attention to a methodological issue in the two studies that may usefully be addressed in such further study. This issue, we suggest, may have tempered the results in boys, resulting in an underestimate of the HRs. In both studies,[1,2] Ing *et al* disregarded all circumcisions performed during the birth admission, treating all neonatally circumcised boys as 'unexposed' (Ing C, personal communication). This is potentially problematic, since 55-64 percent of male neonates in Texas and 61-69 percent of those born in New York underwent circumcision during the birth hospitalization in the study period 1999-2010.[3] By including neonatally circumcised boys in the 'unexposed' category, this reference group was not free of potentially traumatic surgical experiences.

To the extent that early-life surgery *per se* or/and associated traumatic experiences and potential complications account for the observed association with ADHD risk, a possibility admitted by Ing *et al* in both papers,[1,2] the authors' decision to consider boys who underwent neonatal circumcision as 'unexposed' will have produced HR estimates that are too conservative in boys. Accordingly, the higher HR for ADHD among girls than boys undergoing one of the studied surgeries relevant to both sexes (girls: HR, 1.56 (95% CI, 1.38-1.76); boys: HR, 1.37 (95% CI, 1.28-1.47)) might reflect underestimation of the effect among boys by the inclusion of neonatally circumcised boys in the reference category of 'unexposed' boys.

We would encourage Ing *et al* to rerun the following two analyses from their most recent paper[2] and add a new analysis to examine the relevance and potential impact of our above-mentioned concerns.

Firstly, the authors report an overall HR of 1.37 (95% CI, 1.30-1.44) for persistent ADHD medication use after surgeries requiring general anesthesia among boys and girls combined (Figure 3, Panel A). What would this HR be, if neonatally circumcised boys (i.e., those who were circumcised during the birth admission) were excluded, thus restricting the 'unexposed' category to children who truly did not undergo any surgery?

Secondly, in the surgery-specific analysis, the authors report a HR of 1.23 (95% CI, 1.10-1.38) among boys for persistent ADHD medication use after postneonatal circumcision (Figure 3, Panel D). What would this HR be after exclusion of neonatally circumcised boys from the 'unexposed' category?

Thirdly, we kindly ask the authors to calculate a HR for persistent ADHD medication use, comparing hazard rates among boys undergoing postneonatal circumcision (approximately 91% of which required general anesthesia)[4] with the hazard rates among boys undergoing neonatal circumcision (the vast majority of which are carried out without general anesthesia). The latter novel analysis is likely to provide important clues for determining whether it is the surgery *per se* or/and associated traumatic experiences and potential complications, the general anesthesia, or a combination of the two that explain the observed increased ADHD risk among boys who were circumcised after the neonatal period.

References

1. Ing C, Sun M, Olfson M, et al. Age at exposure to surgery and anesthesia in children and association with mental disorder diagnosis. *Anesth Analg* 2017;125:1988-98.

2. Ing C, Ma X, Sun M, et al. Exposure to surgery and anesthesia in early childhood and subsequent use of attention deficit hyperactivity disorder medications. *Anesth Analg* 2020;131:723-33.

3. Owings M, Uddin S, Williams S. Trends in circumcision for male newborns in U.S. hospitals: 1979-2010. National Center for Health Statistics website. https://www.cdc.gov/nchs/data/hestat/circumcision_2013/circumcision_2013.pdf. Accessed October 14, 2020.

4. Wiswell TE, Tencer HL, Welch CA, Chamberlain JL. Circumcision in children beyond the neonatal period. *Pediatrics* 1993;92:791-3.

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